ReMind: An AI-Assisted Mental Health Support Platform Integrating Conversational AI and Professional Psychological Counseling for Students

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Abstract. Mental health issues among students have become increasingly significant due to academic pressure, career uncertainty, and the influence of social media. However, access to professional psychological services is often limited by cost, time constraints, and privacy concerns. This paper presents ReMind, an AI-assisted mental health support platform that combines conversational artificial intelligence with professional psychological counseling services. The platform provides anonymous emotional support, community interaction, appointment scheduling with licensed experts, and administrative management features. The proposed solution aims to improve accessibility, affordability, and effectiveness of mental health support for students.

Keywords: Mental Health · Conversational AI · Psychological Counseling · Student Well-being · Digital Platform.

1 Introduction

Mental health challenges among students have become increasingly prevalent due to academic pressure, career uncertainty, social expectations, and the widespread influence of social media. Despite the growing demand for psychological support, many students face barriers when seeking professional help, including high consultation costs, limited accessibility, scheduling constraints, and concerns regarding privacy and social stigma.

To address these challenges, this paper proposes **ReMind**, an AI-assisted mental health support platform that integrates conversational artificial intelligence with professional psychological counseling services. The platform aims to provide accessible, affordable, and confidential mental health support through AI-powered emotional assistance, community interaction, and online appointment scheduling with licensed psychological experts.

2 Problem Statement

Students frequently experience stress, anxiety, and emotional difficulties resulting from academic workloads, financial concerns, interpersonal relationships, and future career planning. However, existing mental health services often present several limitations:

- High cost of professional counseling services.
- Limited availability of licensed psychologists.
- Reluctance to seek help due to privacy concerns.
- Lack of immediate emotional support during critical situations.
- Absence of integrated platforms combining AI support and professional counseling.

These limitations motivate the development of ReMind as a comprehensive and scalable mental health support solution.

3 Related Work

Several digital mental health platforms have emerged in recent years. Online counseling systems such as BetterHelp and Talkspace provide virtual therapy sessions with licensed professionals. AI-powered solutions such as Woebot and Wysa utilize conversational agents to provide emotional support and self-help guidance.

Although these systems have demonstrated effectiveness in improving accessibility to mental health services, most focus exclusively on either AI-based support or professional counseling. ReMind differentiates itself by integrating both approaches within a single platform while incorporating anonymous participation and community-driven support.

4 Proposed System

ReMind is designed as a web-based platform that connects students, psychological experts, and administrators through a centralized mental health support ecosystem. The platform combines artificial intelligence and professional counseling to provide immediate assistance and long-term psychological support.

4.1 System Architecture

The system follows a multi-layer architecture consisting of:

- **Presentation Layer:** Web-based user interface.
- **Application Layer:** Business logic and service management.
- **AI Service Layer:** Conversational AI support system.
- **Data Layer:** User profiles, appointments, payments, reports, and forum data.

The AI module serves as the first line of emotional support, while professional experts provide in-depth counseling through scheduled appointments.

4.2 User Roles

Table 1. User Roles and Responsibilities

Role	Responsibilities
Guest	Browse expert profiles, view community discussions, register an account, and log in using email/password or Google authentication.
Student	Manage profile information, enable anonymous mode, communicate with the AI chatbot, participate in community forums, schedule appointments, communicate with experts after payment, and submit reports.
Psychological Expert	Manage professional profiles, configure consultation schedules, conduct counseling sessions, export appointment data, and monitor performance reports.
Administrator	Approve expert applications, manage forums, review reports, suspend violating accounts, manage permissions, configure commission rates, and monitor system analytics.

4.3 Core Features

The platform provides a comprehensive suite of features designed to cater to the specific needs of its users. These features are structured around the core objectives of providing immediate emotional support, fostering a safe community, and facilitating professional counseling. Table 2 summarizes these main modules.

AI-Powered Emotional First Aid The platform integrates an AI chatbot that serves as the first point of contact for users experiencing emotional distress. Using large language models (LLMs) and advanced prompt engineering techniques, the backend aggregates recent conversation histories to provide empathetic, immediate, and context-aware responses without requiring the development of a proprietary AI model from scratch.

Anonymous Community Forum with Automated Moderation To encourage open expression without fear of judgment, ReMind features an anonymous forum. To maintain a safe and supportive environment, the system employs an automated keyword-filtering mechanism using Regular Expressions (Regex) at the server level to detect and mask toxic or inappropriate language before it is saved to the database.

Table 2. Core Features of ReMind

Module	Description
AI Chat Support	Provides preliminary emotional support and mental health guidance through conversational artificial intelligence.
Community Forum	Allows users to create posts, share experiences, and interact with other members.
Appointment Booking	Enables students to schedule counseling sessions with licensed psychological experts.
Secure Messaging	Provides communication between students and experts following successful appointment booking.
Expert Management	Supports expert profile management, scheduling, consultation tracking, and reporting.
Administrative Dashboard	Provides moderation functions, financial reporting, and system analytics.

Professional Counseling and Booking Collision Prevention For users requiring specialized intervention, the platform offers a booking system to schedule 1-on-1 sessions with licensed professionals. To ensure operational reliability, the backend implements a strict collision detection algorithm that cross-references new appointment requests with the expert's existing confirmed schedules, preventing double-booking and ensuring seamless expert-patient connections.

5 Implementation and Technologies

The development of ReMind follows a modern, scalable technology stack to ensure high performance, security, and cross-platform availability.

5.1 Technology Stack

The system is implemented using the following core technologies:

- **Frontend (Web & Mobile):** The web application is built using ReactJS, utilizing its Virtual DOM for rapid UI rendering. For broader accessibility, a mobile application is developed with React Native via Expo Go. This allows a single codebase for Android/iOS deployment while sharing a similar ecosystem and core logic with the ReactJS web version.
- **Backend:** Node.js is employed for the backend server due to its event-driven, non-blocking I/O model, making it highly efficient for handling thousands of concurrent connections, which is essential for real-time chat applications.
- **Database:** MongoDB, a flexible NoSQL database, is used for rapid development and scalable storage of user profiles, chat histories, and forum documents.

5.2 Non-Functional Requirements and Security

To ensure a robust and user-centric platform, several non-functional aspects were prioritized:

- **Security & Authorization:** User passwords are securely hashed using bcrypt. Authentication and API route protection are managed via JSON Web Tokens (JWT) coupled with Role-Based Access Control (RBAC) to correctly route Guests, Students, Experts, and Admins.
- **Performance Optimization:** The platform implements database-level pagination (Limit/Offset) for forum feeds to prevent system overload. Furthermore, real-time communication is powered by WebSockets (Socket.io), utilizing isolated 'rooms' corresponding to specific appointment IDs to guarantee message privacy and system performance.
- **Reliability:** Comprehensive error handling (try/catch mechanisms) on the Node.js backend ensures that system errors are caught and returned as user-friendly alerts on the frontend, preventing exposure of raw stack traces.

6 Business Model and Monetization

ReMind adopts a freemium business model to balance accessibility and sustainability.

Table 3. Business Model of ReMind

Component	Description
Free Services	AI chatbot support, expert profile browsing, and participation in community discussions.
Premium Services	One-on-one counseling sessions and direct communication with psychological experts.
Revenue Stream	Commission fees collected from successful counseling appointment transactions.
Target Users	High school students and university students.
Value Proposition	Accessible, affordable, confidential, and technology-enhanced mental health support.

7 Discussion

The integration of conversational AI and professional psychological counseling allows ReMind to provide scalable and efficient mental health support. AI-powered assistance offers immediate responses and reduces the workload of human experts, while licensed professionals address complex psychological issues requiring specialized intervention.

Furthermore, the anonymous participation feature encourages students to seek support without fear of social judgment, thereby increasing engagement and accessibility.

8 Conclusion

This paper presented ReMind, an AI-assisted mental health support platform designed to improve access to psychological services for students. By integrating conversational AI, community engagement, and professional counseling into a unified platform, ReMind addresses critical barriers associated with traditional mental health services.

Future development will focus on enhancing AI response quality, implementing personalized mental health recommendations, and incorporating advanced analytics to support early detection of mental health risks among students.

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